Date: 23.11.2020

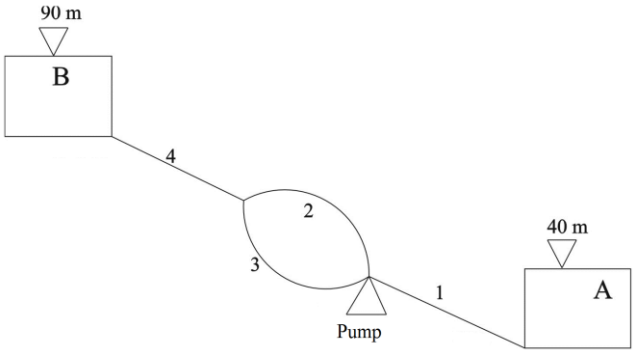
2020-2021 FALL SEMESTER HYDRAULICS COURSE

Name: **Surname:** **Student ID:**

Question 4 (25):

The discharge of 80 l/s is pumped from reservoir A to reservoir B in Figure below. The pipes are made of steel and the Strickler roughness coefficient is k =90. The efficiency of the pump is η= 0.8. According to this;

a) Draw the energy grade line of the system.
b) Calculate the power of the pump.



Where a, and b are the last two digits of your student ID. For example, if the student ID is 12345**67**, **a is 6** and **b is 7**. For this student ID, the lengths are as follows: L_1=167 m, L_2= 376 m, L_3=267 m, L_4= 376 m, D_1= 0.27 m, D_2=0.15 m, D_3= 0.1 m, D_4=0.27 m

	Pipe 1	**Pipe 2**	**Pipe 3**	**Pipe 4**
D (Pipe diameter) (meter)	0.2b	0.15	0.1	0.2b
L (Pipe length) (meter)	1ab	3ba	2ab	3ba